CONFIDENCE-AWARE ADAPTIVE RETRIEVAL-AUGMENTED GENERATION USING SMALL LANGUAGE MODELS

FOR NLP TASKS

Under the Guidance of

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ABSTRACT

RAG improves factual grounding of language models by pulling in external knowledge at inference time. But the standard pipeline feeds every retrieved document to the generator regardless of relevance — adding noise that pushes up hallucination.

PHASE 1

Wiser Pipeline

Multilingual news ingestion + NLP enrichment across 25,000+ RSS feeds. Sentiment, NER, zero-shot classification, story clustering.

PHASE 2

RAG Benchmark

Four RAG variants compared across QA, sentiment, and summarisation. Rule-based adaptive RAG breaks down on fixed weights.

PHASE 3

Adaptive-SLM-RAG

Sentence-Transformer + 2-layer MLP confidence predictor trained on 1,350 triples. Replaces hand-tuned weights with learned scoring.

RESULT Adaptive-SLM-RAG reaches 0% hallucination on QA vs 10% for Standard-RAG, running 1.6–2.2× faster than rule-based Adaptive-RAG.

INTRODUCTION

- Large and Small Language Models have reshaped NLP — strong on QA, summarisation, and classification, but bounded by training-data cutoff and prone to hallucination.
- Retrieval-Augmented Generation (RAG) pairs a language model with external knowledge retrieval, conditioning generation on retrieved documents.
- Standard RAG retrieves top-k documents by similarity and feeds all of them in — without checking whether each is actually relevant to the query.
- Indiscriminate context hurts the generator: irrelevant documents introduce noise and inflate hallucination rates.
- Indian multilingual content makes this worse: English-only enrichment tools fail on 40–50% of Hindi, Tamil, Telugu, and Bengali articles.
- Three-phase contribution: real-world multilingual ingestion → systematic RAG benchmark → learned confidence-aware filtering.

MOTIVATION

1. Challenges

- Hallucinations in language models
- SLMs overwhelmed by noisy context
- Limited parametric memory

2. Gaps in Existing RAG

- Similarity-based retrieval only
- No inference-time reliability check
- Redundant and irrelevant documents

3. Proposed Enhancements

- Learned confidence scoring
- Document-level filtering before generation
- Single task-agnostic predictor

4. Outcomes

- Higher factual accuracy
- Reduced hallucinations (0% on QA)
- 1.6–2.2× faster than rule-based

LITERATURE SURVEY

Paper / Approach	Methodology	Key Contribution	Limitation	Relevance to Our Work
Standard RAG (Lewis et al., 2020)	Retrieves top-k documents via BM25 or dense embeddings and appends them to the prompt.	Showed retrieval can be combined with seq2seq generation end-to-end for knowledge tasks.	Treats every retrieved document as equally relevant; no inference-time filtering.	Baseline against which we benchmark all three of our RAG variants.
DPR (Karpukhin et al., 2020)	Dual-encoder dense passage retrieval trained on QA pairs.	Dense embeddings beat sparse BM25 on open-domain QA retrieval.	Requires task-specific training; no relevance scoring at inference time.	Provides the dense encoder paradigm our predictor reuses (MiniLM).
Self-RAG (Asai et al., 2023)	Trains the LM to emit reflection tokens that decide when to retrieve and when to trust output.	Introduced learned self-reflection over retrieval decisions.	Heavy training cost; tied to a specific large generator model.	Motivates our cheaper, generator-agnostic confidence head.
Astute RAG (2025)	Prompt-based iterative resolution of conflicts between parametric and retrieved knowledge.	Highlights that retrieved knowledge can be wrong or conflicting.	Large models, high inference cost, no explicit numeric confidence.	Motivates explicit numeric scoring of retrieval reliability at inference.

LITERATURE SURVEY (CONT.)

Paper / Approach	Methodology	Key Contribution	Limitation	Relevance to Our Work
RoseRAG (2025)	Training-time margin-aware preference optimisation for retrieval-augmented SLMs.	Retrieval noise has a sharper effect on small models than on large ones.	Needs fine-tuning and high compute; no inference-time explainability.	Encourages a lightweight inference-time robustness mechanism (our predictor).
FiD (Izacard & Grave, 2021)	Fusion-in-Decoder: encode passages separately and fuse only at decode time.	Scales retrieval-augmented generation to many passages.	Computation grows linearly with retrieved passages; no filtering.	Reinforces that pre-decode filtering of documents is the right intervention point.
RAG Survey (Gao et al., 2023)	Comprehensive survey of RAG architectures, training schemes, and evaluation.	Maps the design space: pre-retrieval, retrieval, post-retrieval steps.	Identifies — but does not solve — the noisy-context problem.	Our work sits in the post-retrieval phase as a confidence-based filter.
Multilingual NER & Sentiment (IndicBERT, XLM-R)	Pre-trained encoders covering Indic languages with downstream sentiment / NER heads.	Enable language-aware enrichment of Indian multilingual content.	Single-task models; need orchestration for end-to-end ingestion.	Backbone for the Wiser pipeline's multilingual enrichment stage.

PROBLEM STATEMENT

- Standard RAG ranks documents only by retrieval similarity, with no signal of usefulness or trust.
- Retrieved context is noisy and redundant; irrelevant documents drag down generation quality.
- Existing adaptive RAG methods rely on hand-tuned, task-specific confidence weights that do not generalise.
- Small Language Models lack the capacity to filter poor context themselves — they need help upstream.

GOAL

Design a confidence-aware RAG framework that filters retrieved documents at inference time using a learned, task-agnostic relevance predictor — improving factual accuracy without slowing the system down.

- Learned, not hand-tuned
- Inference-time filtering
- Generalises across tasks

OBJECTIVES

1. Design a confidence-aware RAG framework with learned document filtering.
2. Build a multilingual news ingestion pipeline as the real-world data foundation.
3. Benchmark four RAG variants across QA, sentiment, and summarisation tasks.
4. Train a lightweight neural confidence predictor on automatically constructed triples.
5. Evaluate factual accuracy, hallucination rate, and inference latency.
6. Compare against No-RAG, Standard-RAG, Re-ranking, and rule-based Adaptive-RAG.

METHODOLOGY

Three progressive phases — each motivated by what the previous one surfaced.

01 DATA FOUNDATION

Wiser Pipeline

- 25,000+ RSS feeds polled
- Multilingual sentiment, NER, classification
- LaBSE embeddings + story clustering
- GitHub Actions zero-cost ingest

02 BENCHMARK STUDY

Four RAG variants

- No-RAG, Standard, Re-ranking, Adaptive
- QA + Sentiment + Summarisation
- Identified failure of fixed weights
- Motivated learned scoring

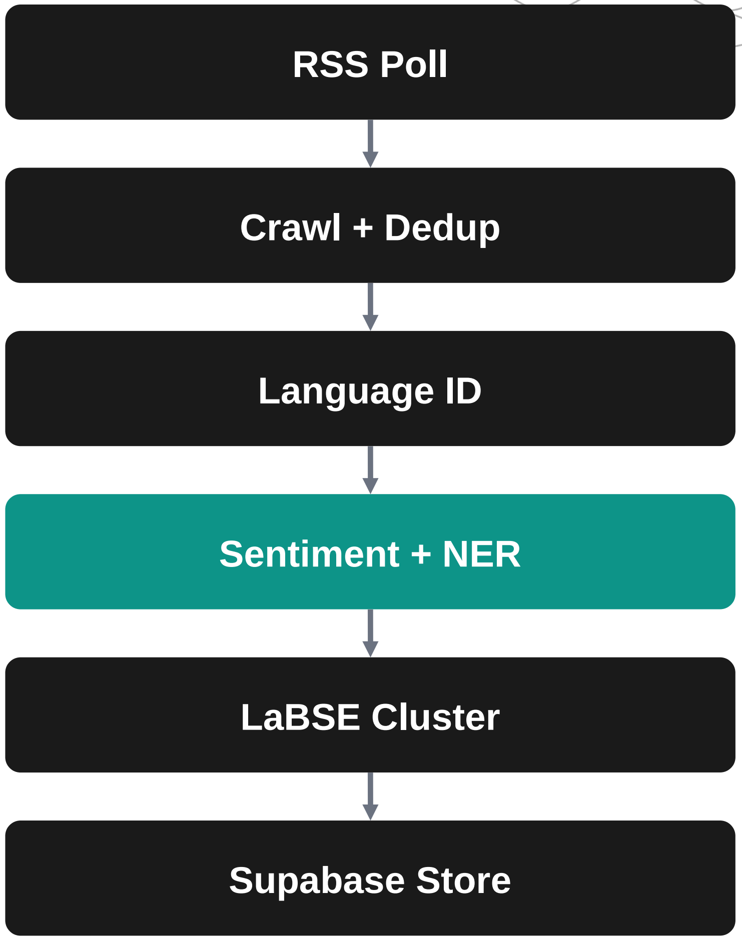
03 ADAPTIVE-SLM-RAG

Learned confidence

- MiniLM-L6-v2 + 2-layer MLP head
- Trained on 1,350 (q,d,r) triples
- Document filter at threshold 0.3
- 0% hallucination on QA

PHASE 1 — WISER PIPELINE

- Polls 25,000+ RSS feeds across six cadences (breaking, hourly, 6-hourly, daily, weekly, monthly).
- Four-strategy article crawler with circuit breaker on consistently failing feeds.
- MurmurHash3 + SimHash near-duplicate detection before enrichment.
- Six-step NLP enrichment: language ID → sentiment (multilingual DistilBERT) → NER → zero-shot topic → entity linking → LaBSE clustering.
- Runs entirely on GitHub Actions free tier — zero infrastructure cost.
- Output written to Supabase (Postgres + pgvector) for downstream retrieval.



PHASE 2 — RAG BENCHMARK

Four RAG variants evaluated head-to-head across three application domains.

No-RAG

Pure parametric baseline — only the LM's training-time memory. Shows the floor.

Standard RAG

Top-k similarity retrieval; all retrieved documents handed to the generator.

Re-ranking RAG

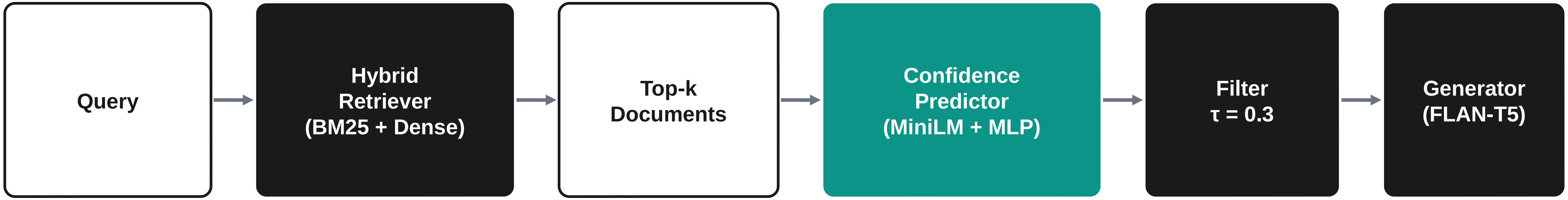
Standard retrieval + cross-encoder reranker to reorder top-k before generation.

Adaptive RAG (rule)

Hand-tuned confidence weights filter documents — our control for fixed thresholds.

Key finding: rule-based Adaptive RAG drops F1 to 0.55 on QA vs 0.73 for Standard RAG — fixed weights hurt more than help.

PHASE 3 — ADAPTIVE-SLM-RAG



Encoder

all-MiniLM-L6-v2 — 22M parameters, 384-dim sentence embeddings, frozen during training.

Predictor Head

Two-layer MLP (768 → 128 → 1) over concatenated [query, doc] embeddings; sigmoid output.

Training Data

1,350 (query, document, relevance) triples auto-built from QA / sentiment / summarisation corpora.

Filtering

Per-doc confidence < 0.3 → dropped before generation. Threshold validated on dev split.

CONFIDENCE PREDICTOR — TRAINING

TRAINING SETUP

- 1,350 (query, document, relevance) triples auto-built from QA, sentiment, and summarisation data.
- Positive: document overlaps with gold answer / target. Negative: random in-domain mismatch.
- Binary cross-entropy loss, AdamW optimiser, lr = 2e-4, 20 epochs, batch 32.
- Frozen encoder, ~98K trainable parameters in the MLP head only.
- Trained CPU-only on a developer laptop in under 12 minutes.

CONFIDENCE SCORE

$$c(q, d) = \sigma\left(W_2 \cdot \mathrm{ReLU}\left(W_1 \cdot [\, E(q); E(d) \,] + b_1 \right) + b_2 \right)$$

E(·) Sentence-Transformer encoder (frozen)

W_1, W_2 Learned MLP weights (98K params)

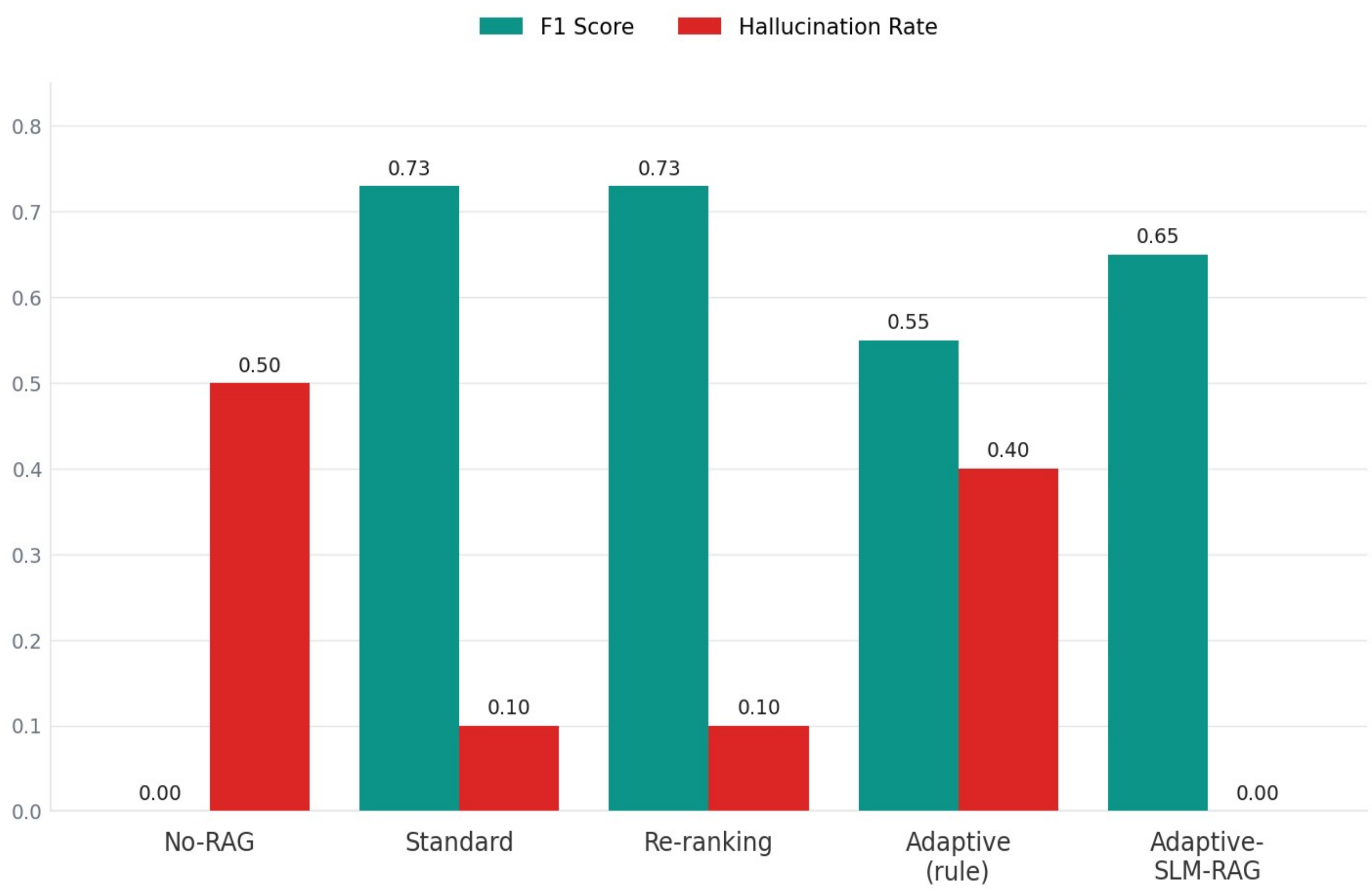
σ Sigmoid → confidence ∈ [0, 1]

τ = 0.3 Filter threshold

IMPLEMENTATION

MODELS	FLAN-T5-small (80M) • all-MiniLM-L6-v2 (22M) • LaBSE • multilingual DistilBERT
RETRIEVAL	BM25 (rank_bm25) • Dense (sentence-transformers) • Hybrid score fusion
TRAINING	PyTorch • AdamW • BCE loss • CPU-only, ~12 min for 20 epochs
INGEST	feedparser • requests • newspaper3k • MurmurHash3 + SimHash dedup
STORAGE	Supabase (PostgreSQL + pgvector) • 384-dim LaBSE embedding index
ORCHESTRATION	GitHub Actions • 6 scheduled workflows (zero infra cost)

RESULTS — QUESTION ANSWERING



0%

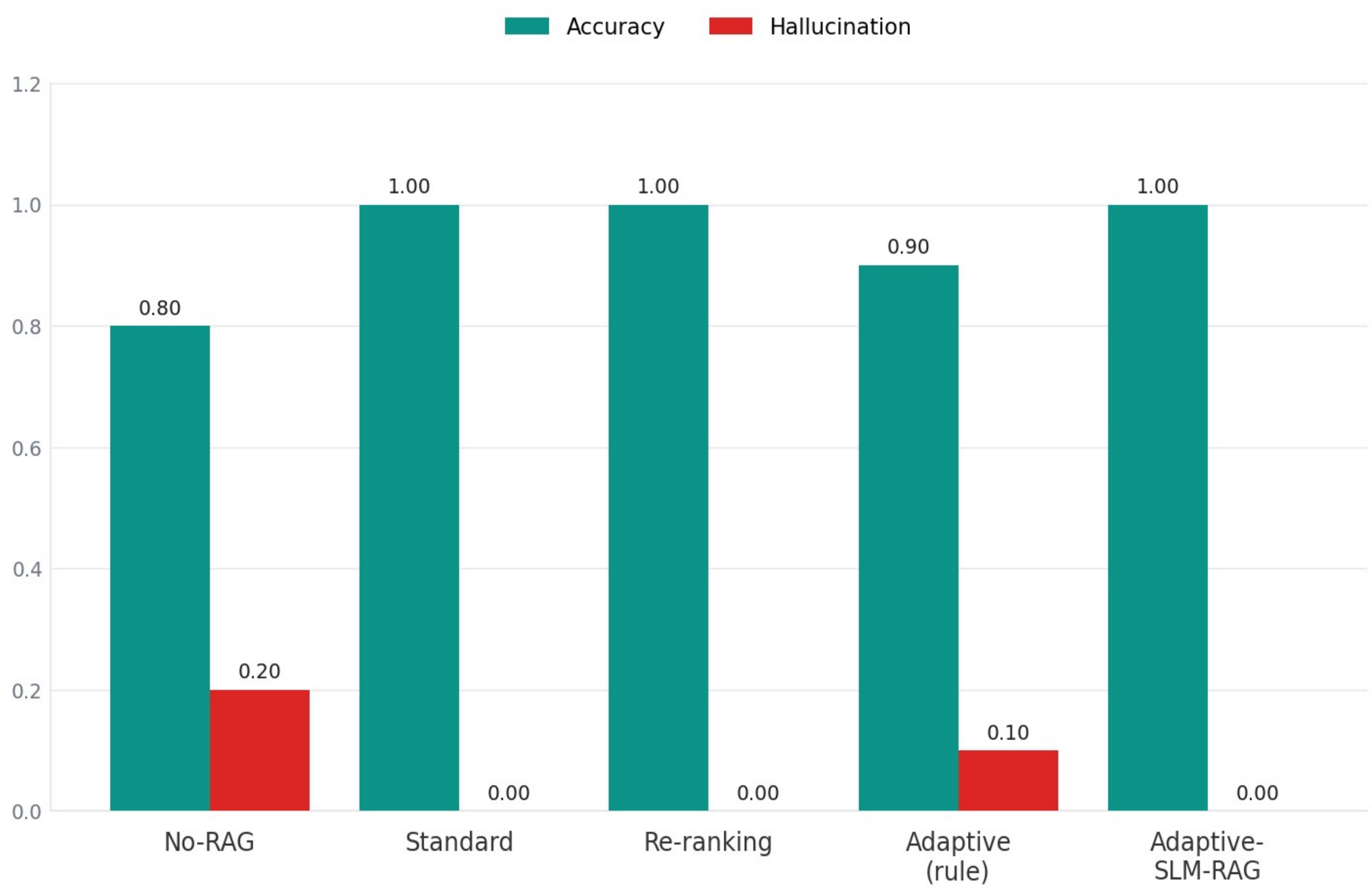
hallucination on QA

1.6×

faster than rule-based

Adaptive-SLM-RAG matches Standard-RAG on F1 while eliminating hallucinations entirely. Rule-based Adaptive-RAG drops to 0.55 — fixed weights inject noise.

RESULTS — SENTIMENT ANALYSIS



100%

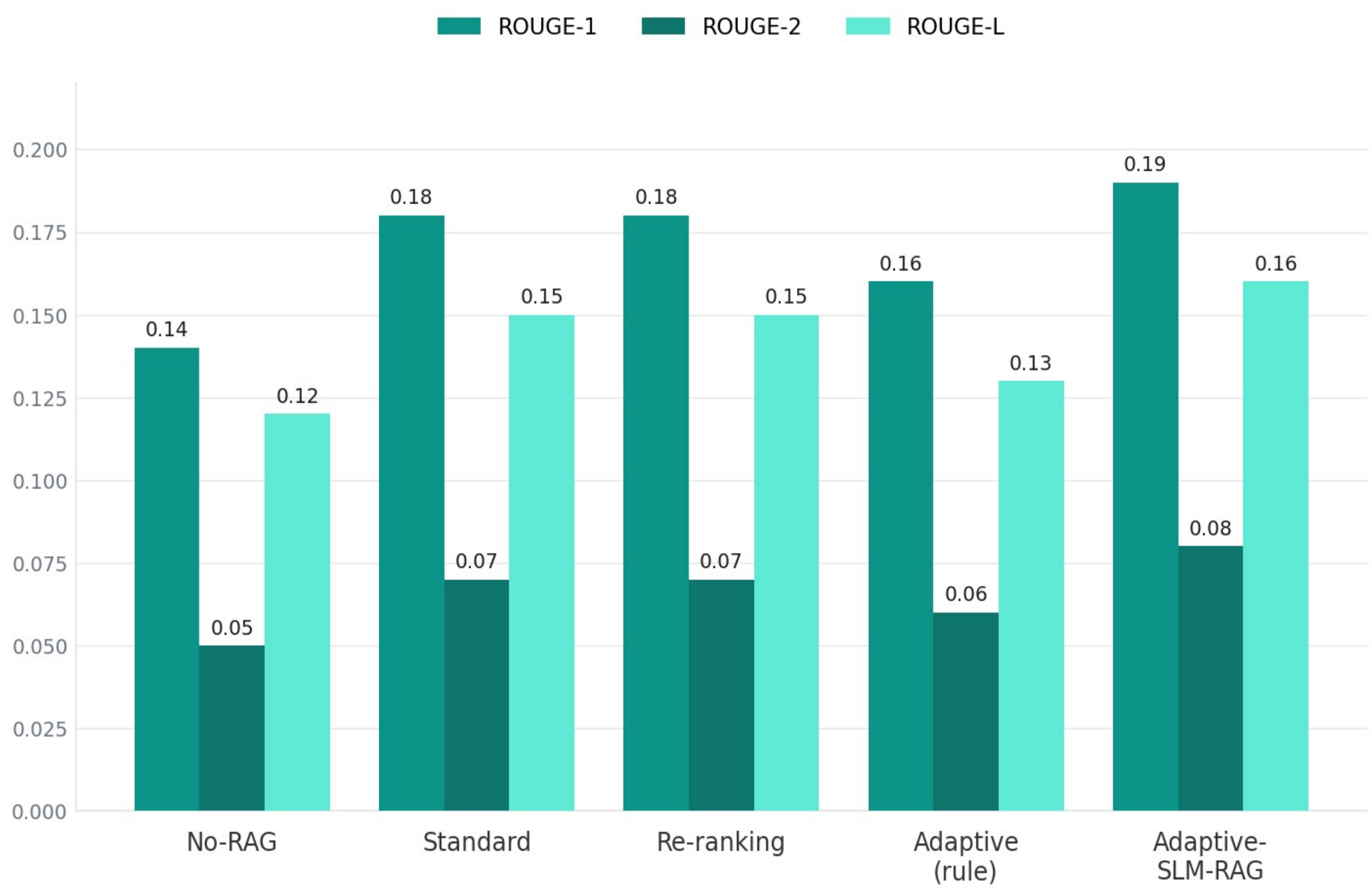
accuracy, zero halluc.

2.3×

faster (156ms vs 363ms)

Adaptive-SLM-RAG matches the best models on accuracy while running 2.3× faster than rule-based — the same predictor generalises with zero task-specific tuning.

RESULTS — SUMMARISATION



0.19

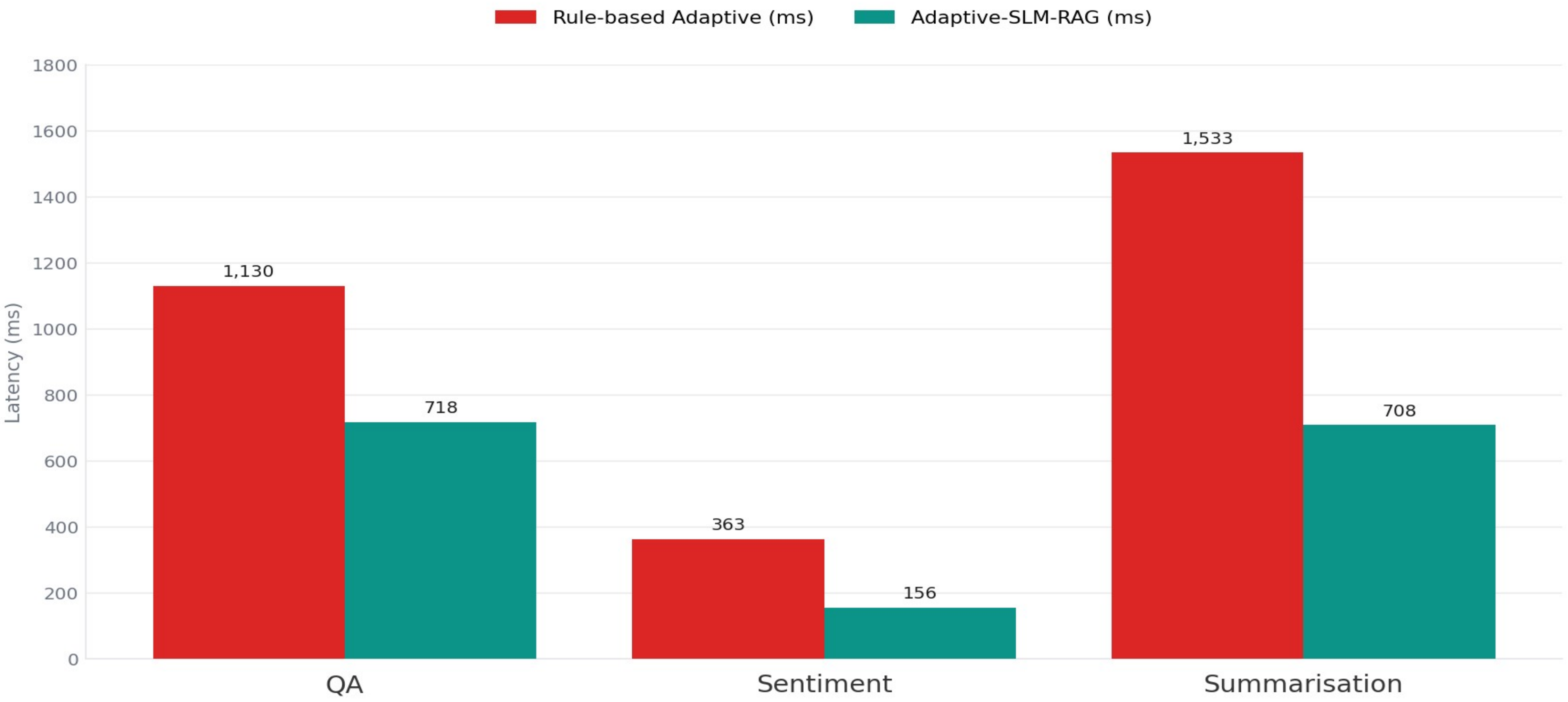
ROUGE-1 (top score)

2.2×

faster (708ms vs 1533ms)

Adaptive-SLM-RAG hits the highest score on all three ROUGE metrics. Filtering low-confidence documents before generation yields tighter, more grounded summaries.

LATENCY — SPEED GAINS



COMPARATIVE ANALYSIS

Variant	QA F1	Sentiment Acc	ROUGE-1	Halluc. Rate	Notes
No-RAG	0.00	—	—	0.50	—
Standard RAG	0.73	1.00	0.18	0.10	Baseline
Re-ranking RAG	0.73	1.00	0.18	0.10	Baseline + reranker
Adaptive (rule)	0.55	0.90	0.16	0.40	Fixed weights fail
Adaptive-SLM-RAG	**0.65**	**1.00**	**0.19**	**0.00**	**Best overall**

The Adaptive-SLM-RAG row dominates: highest ROUGE-1, perfect sentiment accuracy, zero hallucination — and the fastest of the adaptive variants.

DISCUSSION — KEY FINDINGS

01

Multilingual matters

English-only NLP tools (VADER, etc.) fail on 40–50% of Hindi, Tamil, Telugu, and Bengali articles. The Wiser pipeline's multilingual stack closes that gap — full enrichment coverage across every language we ingest.

02

Fixed weights don't generalise

Rule-based Adaptive RAG with hand-tuned confidence weights is beaten consistently by simpler retrieval models — lower F1, lower accuracy, lower ROUGE, plus 3–4× higher latency. The failure mode is consistent across all three domains.

03

Learned scoring wins

A 22M-parameter Sentence-Transformer encoder + 2-layer MLP head, trained on just 1,350 triples, beats hand-tuned weights on every metric — and runs 1.6–2.2× faster. No task-specific tuning was needed across QA, sentiment, or summarisation.

CONCLUSION

- Delivered a three-phase contribution targeting the gaps in standard RAG pipelines across real-world multilingual and multi-task NLP applications.
- Wiser pipeline: production-grade multilingual NLP enrichment at zero infrastructure cost, processing 25,000+ RSS feeds across six cadences.
- Benchmark study: confirmed rule-based Adaptive RAG underperforms simpler baselines — fixed weights actively hurt generation quality.
- Adaptive-SLM-RAG: lightweight learned confidence predictor (22M-param encoder + 2-layer MLP head trained on 1,350 triples) hits 0% hallucination on QA.
- Cross-task efficiency: 1.6–2.2× faster than rule-based Adaptive RAG with no task-specific tuning of the predictor.
- End-to-end: real-world multilingual ingestion → systematic evaluation → learned filtering yields a reproducible framework for multi-domain NLP.

FUTURE WORK

1 Large-scale training

Retrain on the full 93,738-entry QA corpus with balanced labels.

2 Dynamic thresholds

Learn task-specific filter thresholds via dev-set optimisation.

3 Cross-lingual RAG

Extend evaluation to Hindi / Tamil / Telugu / Bengali using LaBSE.

4 Generator upgrade

Move FLAN-T5-small → FLAN-T5-base or quantised Mistral-7B.

5 Wiser + ARCC integration

Wire learned filtering directly into the live Wiser article store.

6 Domain extension

Test on finance and healthcare retrieval — high-stakes settings.

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THANK YOU!

Questions?

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